

nlmixr²: past, present and future

Matthew Fidler

On behalf of the nlmixr² team

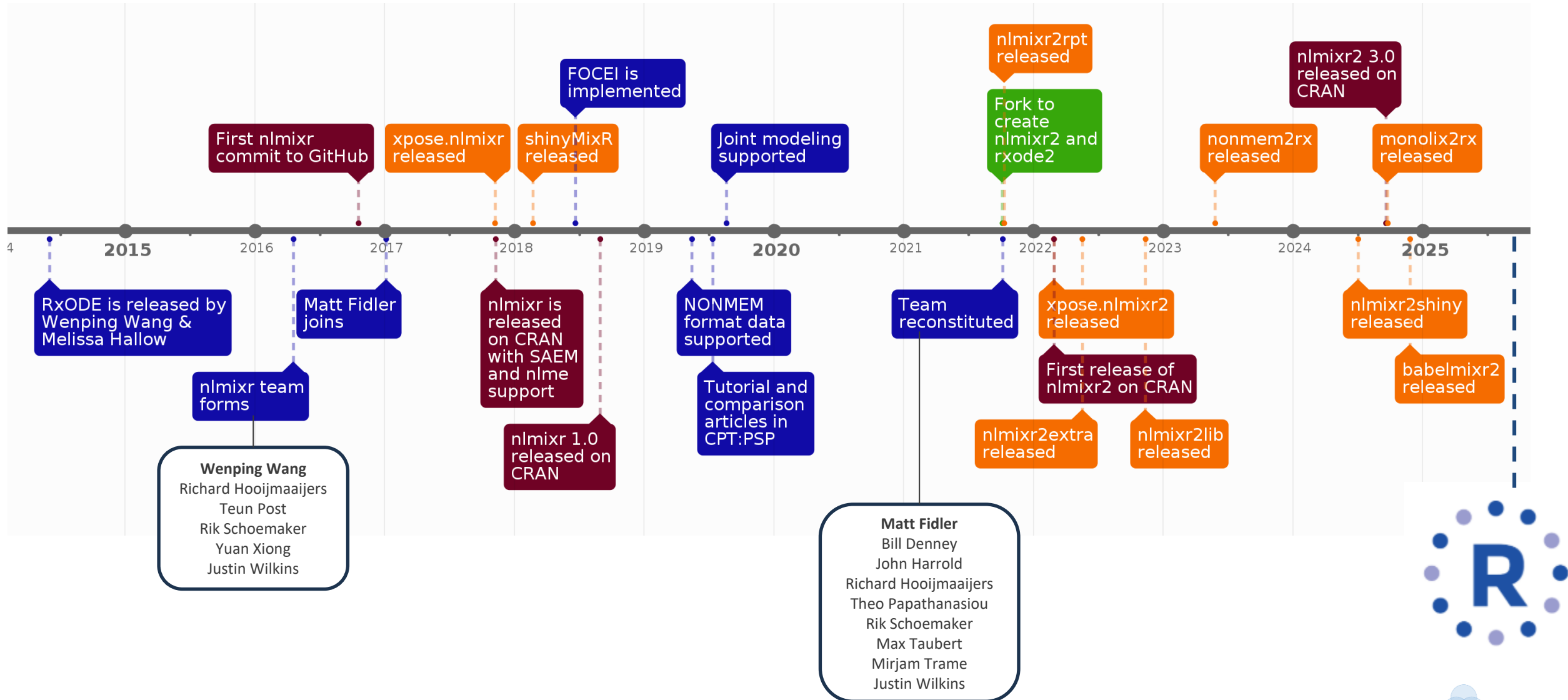


Agenda

Topic	Presenter	Time (ET)
Introduction	Matt Fidler	9:00 AM
Intro to nlmixr2/rxode2 in pharmacometrics analyses	Anne Keunecke	9:10 AM – 10:10 AM
Coffee break		10:10 – 10:25
Simulation approach (Study design)	Anne Keunecke + nlmixr2 WG	10:25 AM ET – 11:25 AM ET
Survival Analysis in nlmixr2 ecosystem	Justin Wilkins	11:25 AM to 12:00 PM
AI-assisted pharmacometric analysis	Mutaz Jaber	12:00 PM – 12:55 PM
Future and closing remarks – New features and pre-releases	Matt Fidler	12:55 PM to 1:00 PM

nlmixr² evolution

nlmixr2 evolved over 20 years, from code first written by Wenping Wang in 2005 to RxODE, which became the foundation for nlmixr in 2016. Renamed nlmixr2 in 2021, it now has a broad package ecosystem and a strong user base in academia and industry.



The nlmixr² team

The nlmixr2 R Consortium Working Group is an international group from **nine large pharmaceutical** companies and CROs. Team members' time is donated by their organizations.

 **NOVARTIS**

CERTARA[®]

Genentech

A Member of the Roche Group

 **GILEAD**

GSK

 **HUMAN
PREDICTIONS**

LAPP Leiden
Advanced PK-PD

OCCAMS

METRUM
RESEARCH GROUP

 **Pfizer**

 **nlmixr**



The nlmixr2 ecosystem is growing with contributions in and out of the working group

nlmixr² team structure & governance structure

Advisory board

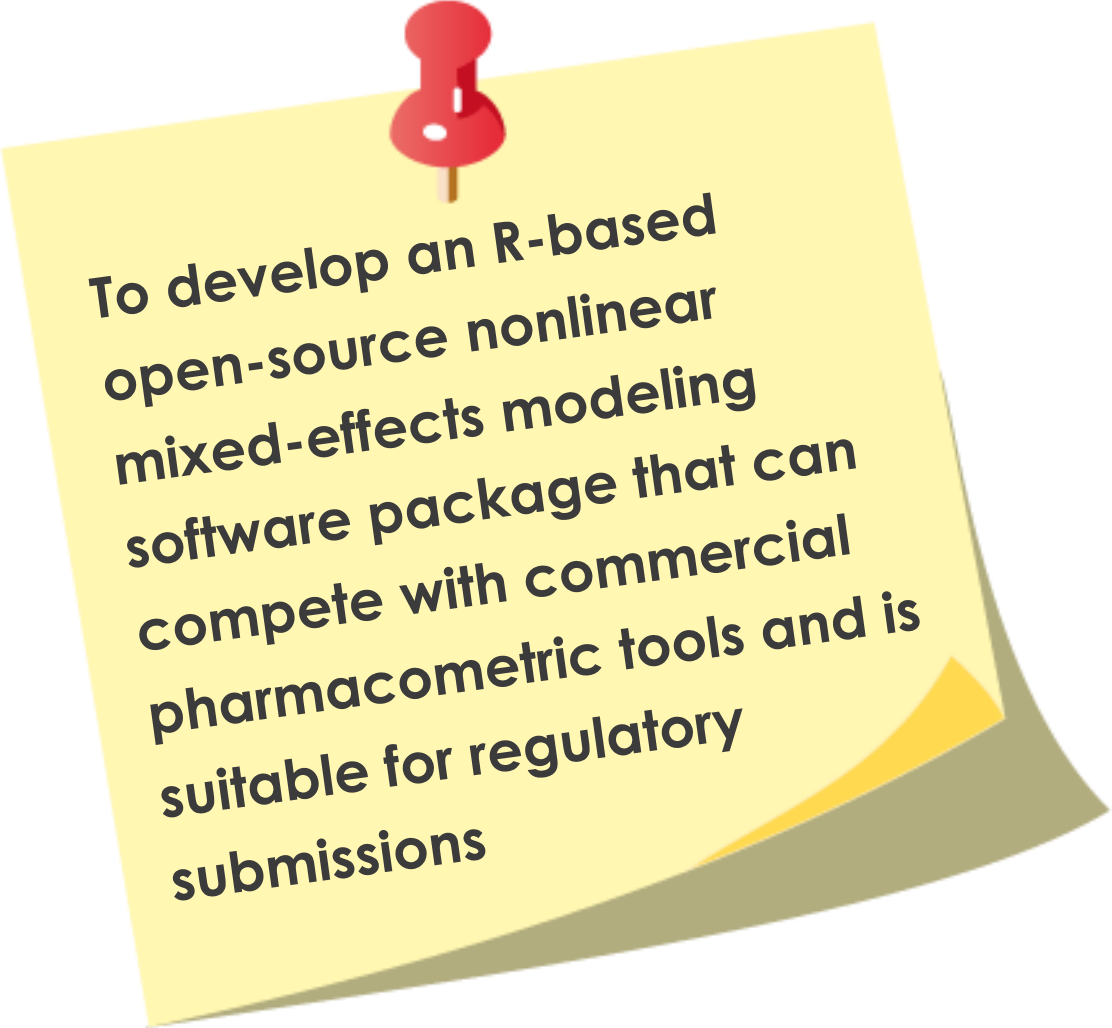
- Guides development priorities
- Provides sustainability input
- Advises on direction
- Meets biannually or as-needed

Development & Governance

- A R-Consortium Working group
- Development is led by **Matt Fidler** with the majority of core contributions
- Other team members provide:
 - Core tool coding/support
 - Satellite package support
 - Training & Documentation
 - Testing

Advisory board members

- **Paolo Denti**
University of Cape Town (South Africa)
- **Jeffry Florian**
FDA (USA)
- **Marc Gastonguay**
Metrum (USA)
- **Irina Baltcheva**
Novartis, (Switzerland)
- **Manuel Ibarra**
Universidad de la República (Uruguay)
- **Mats Karlsson**
Uppsala University (Sweden)
- **Joe Standing**
University College London, (UK)



To develop an R-based
open-source nonlinear
mixed-effects modeling
software package that can
compete with commercial
pharmacometric tools and is
suitable for regulatory
submissions

Thank you

